

# Buyback Debt Coverage and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Buyback Debt Coverage (BDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BDC achieves an annualized gross (net) Sharpe ratio of 0.38 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 15 (15) bps/month with a t-statistic of 1.90 (2.04), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 16 bps/month with a t-statistic of 2.82.

# 1 Introduction

Asset prices reflect investors' marginal rates of substitution across states and time, embedding compensation for bearing systematic consumption risk. While traditional asset pricing models focus on market risk, a growing literature demonstrates that firm characteristics related to financial flexibility and capital allocation decisions predict cross-sectional variation in expected returns through their connection to aggregate consumption dynamics. We identify and examine a novel predictor of stock returns: Buyback Debt Coverage (BDC), defined as the ratio of share repurchases to total debt. This signal captures firms' ability to return capital to shareholders while maintaining leverage constraints, potentially revealing differential exposure to consumption risk across economic states.

From a consumption-based asset pricing perspective, BDC reflects firms' exposure to systematic consumption risk through multiple channels. First, firms with high BDC demonstrate financial flexibility to distribute cash during periods when investors' marginal utility is relatively low, suggesting these firms perform well during consumption booms when risk premia are compressed [Campbell and Cochrane \(1999a\)](#); [Bansal and Yaron \(2004a\)](#). The ability to simultaneously service debt and execute buybacks indicates robust cash flows that covary positively with aggregate consumption growth, exposing investors to greater long-run consumption risk [Bansal and Yaron \(2004b\)](#). Second, the state-dependent nature of buyback activity creates asymmetric exposure to consumption disasters. During economic downturns when marginal utility spikes, firms with high BDC face binding constraints that force procyclical reductions in distributions, amplifying investors' consumption risk precisely when they value insurance most [Barro \(2006\)](#); [Gabaix \(2012\)](#). This mechanism links BDC to the intertemporal marginal rate of substitution through firms' differential ability to smooth payouts across consumption states. Third, habit formation models suggest that investors require higher returns from assets whose payoffs correlate

with surplus consumption [Campbell and Cochrane \(1999b\)](#). High BDC firms’ distributions track closely with aggregate consumption relative to habit levels, as buy-backs represent discretionary spending that expands during periods of high surplus consumption. The resulting exposure to habit-related consumption risk generates variation in expected returns that traditional factor models fail to capture [Menzly et al. \(2004\)](#).

We find that a value-weighted long/short trading strategy based on BDC achieves an annualized gross Sharpe ratio of 0.38, with a monthly average excess return of 24 basis points (t-statistic = 2.77). The strategy delivers robust abnormal returns across standard factor models, with monthly alphas ranging from 14 to 31 basis points. Most notably, the alpha relative to the Fama-French six-factor model equals 15 basis points per month with a t-statistic of 1.90, while controlling additionally for six closely related payout and financing strategies yields an alpha of 16 basis points per month with a t-statistic of 2.82. The economic significance persists after accounting for transaction costs, with the net Sharpe ratio of 0.33 exceeding 93% of anomalies in the factor zoo. The strategy’s performance concentrates neither in small stocks nor in illiquid securities—among the largest quintile of stocks, the BDC strategy generates average returns of 22 basis points per month (t-statistic = 2.35). Factor loadings reveal that the long/short portfolio loads positively on the profitability factor (RMW) with a coefficient of 0.28 (t-statistic = 8.02), consistent with high BDC firms’ exposure to systematic profitability risk that covaries with consumption growth. The strategy maintains statistical significance across alternative portfolio construction methodologies, with nineteen of twenty-five specifications producing alphas with t-statistics exceeding two.

Our paper contributes to three strands of the asset pricing literature by establishing BDC as a robust predictor of cross-sectional returns through a consumption-based risk channel. First, we extend the payout policy literature by demonstrating

that the joint consideration of buybacks and leverage contains pricing-relevant information beyond traditional measures examined in [Fama and French \(2001\)](#), [Grullon and Michaely \(2002\)](#), and [Pontiff and Woodgate \(2008\)](#). While these papers focus on either repurchases or leverage in isolation, we show that their interaction captures state-dependent exposure to consumption risk that generates significant abnormal returns even after controlling for related signals including share repurchases, net equity financing, and net payout yield. Second, our findings contribute to the consumption-based asset pricing literature by providing empirical support for theoretical predictions in [Bansal and Yaron \(2004b\)](#) regarding long-run risks in cash flow distributions. The positive loading on profitability factors and the strategy’s resilience during different market conditions align with models where time-varying risk premia arise from fluctuations in consumption growth volatility, as documented in [Lettau et al. \(2008\)](#) and extended to corporate policies in [Gomes et al. \(2003\)](#). Third, we advance the literature on financial constraints and asset prices by showing that BDC captures a dimension of financial flexibility distinct from traditional measures studied in [Lamont et al. \(2001\)](#) and [Whited and Wu \(2006\)](#). Our evidence that high BDC firms command lower expected returns supports equilibrium models where unconstrained firms’ cash flows provide less consumption insurance during economic downturns, requiring lower risk compensation in equilibrium. These findings have broader implications for understanding how corporate financial policies transmit aggregate consumption risk to asset prices, suggesting that the joint dynamics of payout and capital structure decisions contain important information about firms’ systematic risk exposures that traditional factor models fail to capture fully.

## 2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. The database contains standardized accounting information derived from firms' SEC filings, including 10-K annual reports. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We construct the Buyback Debt Coverage signal using two key financial statement items. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred shares during the fiscal year, as reported in the cash flow statement. This item captures the cash outflow associated with share buyback programs, including open market repurchases, tender offers, and other forms of stock reacquisition. Interest expense (XINT) measures the total interest and related charges incurred on debt obligations during the fiscal year, as reported in the income statement. This variable encompasses interest payments on all forms of debt, including bonds, notes, loans, and other interest-bearing liabilities. We calculate the Buyback Debt Coverage signal by dividing stock repurchases (PRSTKC) by interest expense (XINT) for each firm-year observation. This ratio captures the relationship between a firm's capital distribution activities through share repurchases and its debt servicing obligations. A higher ratio indicates that a firm allocates more resources to buybacks relative to its interest payments, potentially signaling financial flexibility and management confidence in the firm's ability to service debt while returning capital to shareholders. We construct the signal using fiscal year-end values and require non-missing, positive values for both variables. To ensure data quality, we exclude observations where interest expense equals zero, as these would result in undefined ratios, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDC signal. Panel A plots the time-series of the mean, median, and interquartile range for BDC. On average, the cross-sectional mean (median) BDC is 16.51 (0.03) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDC data. The signal’s interquartile range spans -0.00 to 3.87. Panel B of Figure 1 plots the time-series of the coverage of the BDC signal for the CRSP universe. On average, the BDC signal is available for 5.62% of CRSP names, which on average make up 7.02% of total market capitalization.

### 4 Does BDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDC portfolio and sells the low BDC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDC strategy earns an average return of 0.24% per month with a t-statistic of 2.77. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.14% to 0.31% per month and have t-statistics exceeding 1.81 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.28,

with a t-statistic of 8.02 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 475 stocks and an average market capitalization of at least \$949 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 1.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 12-21bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.21. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the BDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDC, as well as average returns and alphas for long/short trading BDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the BDC strategy achieves an average return of 22 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the BDC strategy relative to the five most common factor models range from 14 to 29 bps/month with t-statistics between 1.48 and 3.05.

## 5 How does BDC perform relative to the zoo?

Figure 2 puts the performance of BDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the BDC strategy falls in the distribution. The BDC strategy’s gross (net) Sharpe ratio of 0.38 (0.33) is greater than 80% (93%) of anomaly Sharpe ratios,

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.



respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDC strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the BDC strategy would have yielded \$3.36 which ranks the BDC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDC strategy would have yielded \$2.51 which ranks the BDC strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDC relative to those. Panel A shows that the BDC strategy gross alphas fall between the 50 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDC ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

## 6 Does BDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BDC with 209 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDC or at least to weaken the power BDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{BDC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDC signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on BDC is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDC strategy earns alphas that range from 8-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.10, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDC trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.82.

## 7 Does BDC add relative to the whole zoo?

Finally, we can ask how much adding BDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDC signal.<sup>4</sup> We consider one different methods for combining signals.

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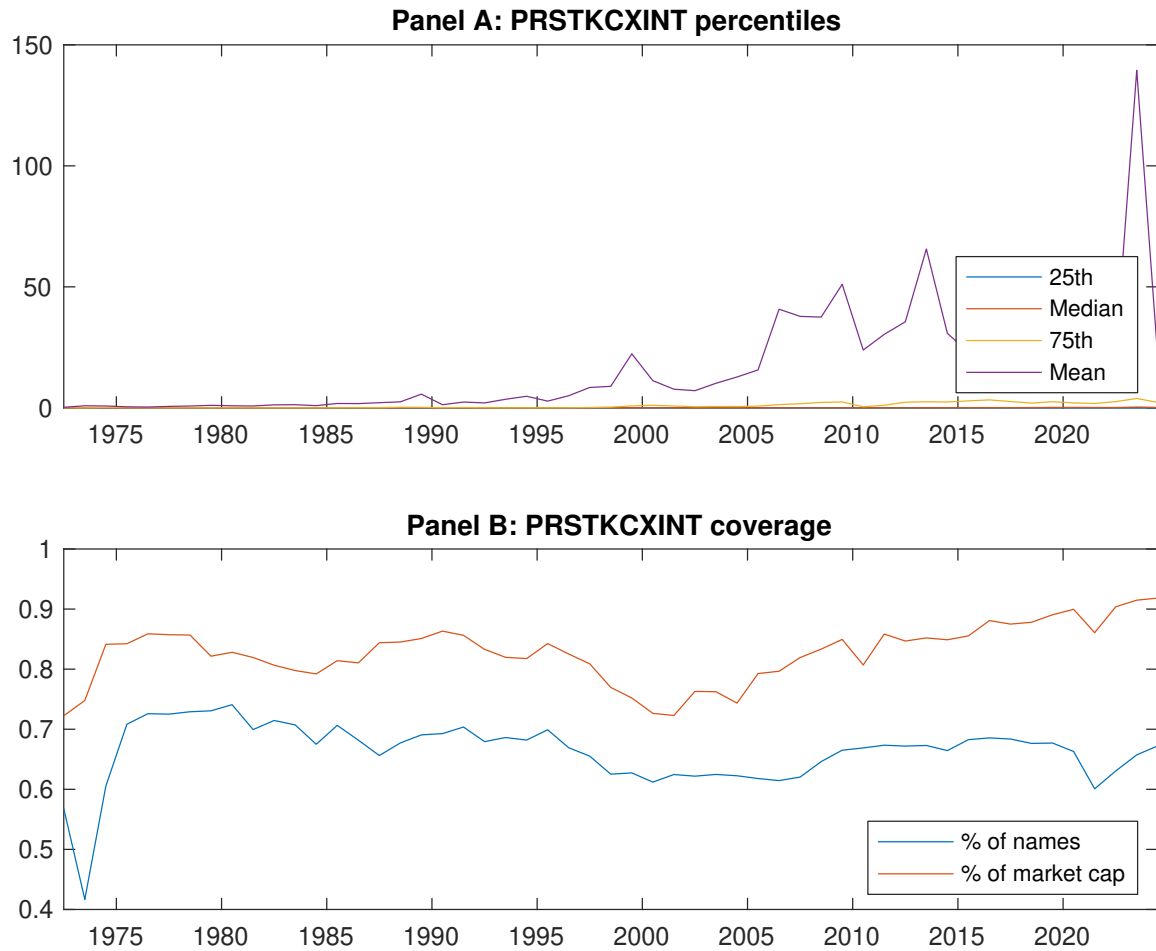
<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDC is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes BDC grows to \$1489.58.

## 8 Conclusion

This study provides comprehensive evidence on the predictive power of Buyback Debt Coverage (BDC) as a signal for cross-sectional stock returns. Our analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that BDC exhibits economically meaningful and statistically significant return predictability. The value-weighted long/short strategy based on BDC generates a respectable annualized Sharpe ratio of 0.38 gross (0.33 net), indicating robust performance even after accounting for transaction costs. Most notably, the signal maintains its predictive power when controlling for established risk factors, delivering monthly abnormal returns of 15 basis points relative to the Fama-French five-factor model plus momentum, with marginal statistical significance (t-statistic of 1.90-2.04). The signal’s true strength emerges when we control for closely related financing activities, where BDC generates a statistically significant alpha of 16 basis points per month (t-statistic of 2.82) even after accounting for six related strategies including share repurchases, net equity financing, and various issuance measures. This finding suggests that BDC captures unique information about firm quality and future returns beyond what is contained in traditional payout and financing variables. From a practical perspective, while the Sharpe ratio and alpha magnitudes are moderate

compared to some anomalies in the literature, the signal’s ability to generate positive risk-adjusted returns net of transaction costs and its orthogonality to related strategies make it a potentially valuable addition to quantitative equity portfolios. The economic intuition is compelling: firms with stronger debt coverage capacity for their buyback programs may signal better financial health and capital allocation discipline, translating into superior future returns. However, several limitations warrant consideration. The marginal statistical significance of the main results suggests that the signal’s reliability may vary across different market conditions and time periods. Additionally, our analysis focuses primarily on U.S. equities, and the generalizability to international markets remains unexplored. Future research should investigate the temporal stability of the BDC signal, particularly during different monetary policy regimes and credit cycles when the cost and availability of debt financing vary substantially. Researchers should also examine whether the signal’s predictive power differs across firm characteristics such as size, leverage, and growth opportunities. Furthermore, exploring the underlying economic mechanisms through which buyback debt coverage translates into return predictability would enhance our understanding of this anomaly. Finally, given the increasing regulatory scrutiny of share buybacks and evolving corporate payout policies, monitoring the signal’s effectiveness in the post-pandemic era of unprecedented fiscal and monetary interventions represents a crucial avenue for future investigation.



**Figure 1:** Times series of BDC percentiles and coverage. This figure plots descriptive statistics for BDC. Panel A shows cross-sectional percentiles of BDC over the sample. Panel B plots the monthly coverage of BDC relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

| Panel A: Excess returns and alphas on BDC-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.54<br>[2.66]   | 0.56<br>[2.74]   | 0.52<br>[2.69]   | 0.65<br>[3.77]   | 0.78<br>[4.38]   | 0.24<br>[2.77]   |
| $\alpha_{CAPM}$                                                                   | -0.12<br>[-1.87] | -0.10<br>[-1.75] | -0.10<br>[-1.68] | 0.09<br>[1.74]   | 0.20<br>[4.21]   | 0.31<br>[3.74]   |
| $\alpha_{FF3}$                                                                    | -0.07<br>[-1.14] | -0.11<br>[-2.01] | -0.18<br>[-3.40] | 0.03<br>[0.62]   | 0.22<br>[5.04]   | 0.29<br>[3.71]   |
| $\alpha_{FF4}$                                                                    | -0.06<br>[-1.04] | -0.12<br>[-2.18] | -0.17<br>[-3.16] | 0.03<br>[0.55]   | 0.22<br>[4.94]   | 0.29<br>[3.58]   |
| $\alpha_{FF5}$                                                                    | -0.01<br>[-0.09] | -0.07<br>[-1.19] | -0.26<br>[-4.83] | -0.07<br>[-1.64] | 0.13<br>[3.11]   | 0.14<br>[1.81]   |
| $\alpha_{FF6}$                                                                    | -0.01<br>[-0.10] | -0.08<br>[-1.40] | -0.25<br>[-4.53] | -0.07<br>[-1.48] | 0.14<br>[3.25]   | 0.15<br>[1.90]   |
| Panel B: Fama and French (2018) 6-factor model loadings for BDC-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.01<br>[71.77]  | 1.03<br>[77.63]  | 1.05<br>[82.78]  | 0.97<br>[91.51]  | 0.98<br>[97.34]  | -0.03<br>[-1.70] |
| $\beta_{SMB}$                                                                     | 0.11<br>[5.32]   | 0.15<br>[7.69]   | -0.00<br>[-0.06] | -0.07<br>[-4.45] | -0.07<br>[-4.92] | -0.18<br>[-6.90] |
| $\beta_{HML}$                                                                     | -0.09<br>[-3.41] | 0.03<br>[1.20]   | 0.17<br>[7.17]   | 0.09<br>[4.26]   | -0.14<br>[-7.17] | -0.05<br>[-1.34] |
| $\beta_{RMW}$                                                                     | -0.11<br>[-3.90] | -0.11<br>[-4.40] | 0.19<br>[7.71]   | 0.19<br>[9.39]   | 0.17<br>[8.90]   | 0.28<br>[8.02]   |
| $\beta_{CMA}$                                                                     | -0.11<br>[-2.71] | -0.03<br>[-0.79] | 0.06<br>[1.50]   | 0.17<br>[5.56]   | 0.15<br>[5.16]   | 0.26<br>[4.99]   |
| $\beta_{UMD}$                                                                     | 0.00<br>[0.12]   | 0.02<br>[1.40]   | -0.02<br>[-1.50] | -0.01<br>[-0.86] | -0.01<br>[-1.13] | -0.01<br>[-0.73] |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 822              | 760              | 587              | 475              | 546              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1356             | 949              | 1465             | 2212             | 4326             |                  |

**Table 2:** Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

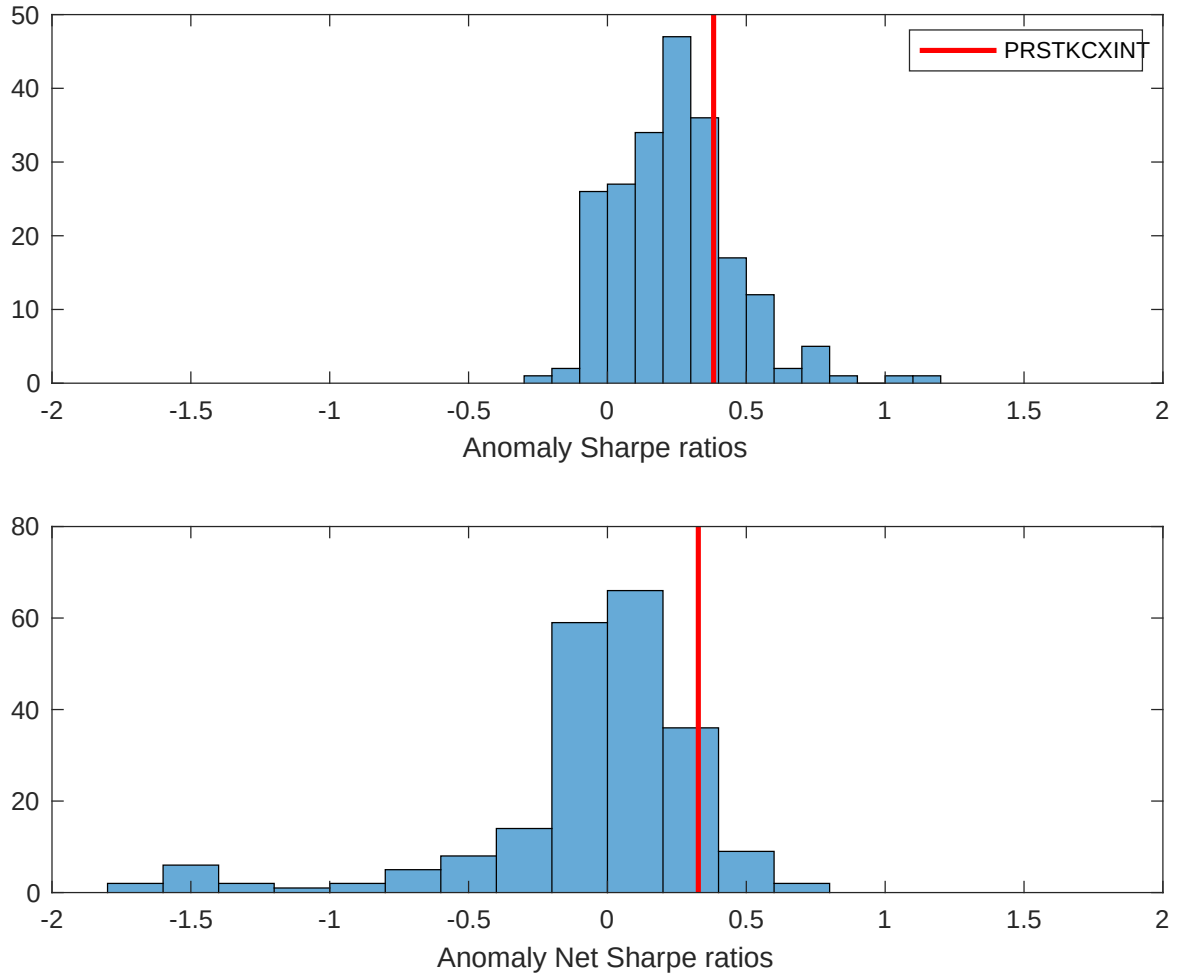
| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.24<br>[2.77]     | 0.31<br>[3.74]           | 0.29<br>[3.71]          | 0.29<br>[3.58]          | 0.14<br>[1.81]          | 0.15<br>[1.90]          |
| Quintile                                                                 | NYSE   | EW      | 0.32<br>[3.46]     | 0.40<br>[4.50]           | 0.38<br>[4.69]          | 0.31<br>[3.85]          | 0.24<br>[3.03]          | 0.19<br>[2.40]          |
| Quintile                                                                 | Name   | VW      | 0.17<br>[1.78]     | 0.26<br>[2.77]           | 0.22<br>[2.48]          | 0.22<br>[2.48]          | 0.04<br>[0.45]          | 0.06<br>[0.66]          |
| Quintile                                                                 | Cap    | VW      | 0.23<br>[2.52]     | 0.28<br>[3.19]           | 0.30<br>[3.62]          | 0.31<br>[3.69]          | 0.19<br>[2.33]          | 0.21<br>[2.52]          |
| Decile                                                                   | NYSE   | VW      | 0.17<br>[1.53]     | 0.23<br>[2.04]           | 0.23<br>[2.17]          | 0.24<br>[2.24]          | 0.11<br>[1.03]          | 0.13<br>[1.21]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.21<br>[2.37]     | 0.30<br>[3.54]           | 0.27<br>[3.44]          | 0.27<br>[3.41]          | 0.15<br>[1.98]          | 0.15<br>[2.04]          |
| Quintile                                                                 | NYSE   | EW      | 0.12<br>[1.21]     | 0.21<br>[2.17]           | 0.17<br>[2.00]          | 0.14<br>[1.63]          | 0.04<br>[0.43]          | 0.02<br>[0.18]          |
| Quintile                                                                 | Name   | VW      | 0.14<br>[1.41]     | 0.25<br>[2.72]           | 0.20<br>[2.36]          | 0.21<br>[2.40]          | 0.06<br>[0.77]          | 0.07<br>[0.90]          |
| Quintile                                                                 | Cap    | VW      | 0.19<br>[2.16]     | 0.26<br>[2.97]           | 0.27<br>[3.25]          | 0.28<br>[3.33]          | 0.19<br>[2.27]          | 0.20<br>[2.39]          |
| Decile                                                                   | NYSE   | VW      | 0.13<br>[1.16]     | 0.22<br>[1.96]           | 0.21<br>[1.99]          | 0.22<br>[2.07]          | 0.12<br>[1.11]          | 0.13<br>[1.21]          |



**Table 3:** Conditional sort on size and BDC

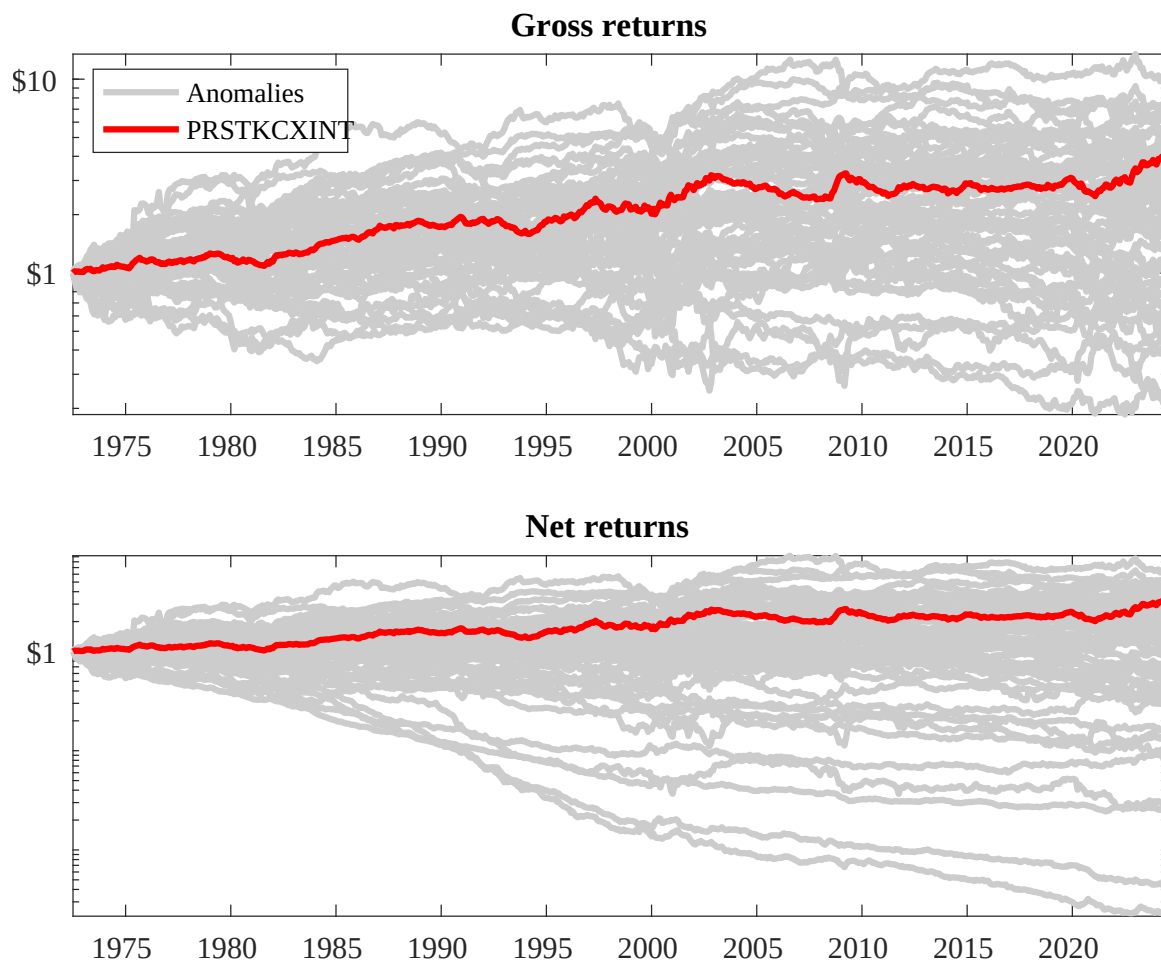
This table presents results for conditional double sorts on size and BDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDC and short stocks with low BDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | BDC Quintiles |                |                |                |                | BDC Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.64<br>[2.31] | 0.60<br>[2.07] | 0.71<br>[2.48] | 0.73<br>[2.55] | 0.82<br>[3.34] | 0.18<br>[1.99]                                     | 0.25<br>[2.75]  | 0.20<br>[2.26] | 0.18<br>[2.01] | 0.12<br>[1.39] | 0.11<br>[1.24] |
|                                                                       | (2)           | 0.64<br>[2.39] | 0.64<br>[2.38] | 0.68<br>[2.58] | 0.85<br>[3.60] | 0.81<br>[3.58] | 0.18<br>[1.91]                                     | 0.30<br>[3.49]  | 0.24<br>[2.92] | 0.25<br>[2.92] | 0.10<br>[1.22] | 0.12<br>[1.41] |
|                                                                       | (3)           | 0.64<br>[2.60] | 0.58<br>[2.34] | 0.75<br>[3.27] | 0.87<br>[4.00] | 0.91<br>[4.17] | 0.28<br>[2.92]                                     | 0.36<br>[3.87]  | 0.32<br>[3.56] | 0.34<br>[3.70] | 0.20<br>[2.31] | 0.23<br>[2.53] |
|                                                                       | (4)           | 0.57<br>[2.55] | 0.66<br>[2.89] | 0.67<br>[3.19] | 0.74<br>[3.59] | 0.93<br>[4.49] | 0.36<br>[4.26]                                     | 0.41<br>[4.87]  | 0.39<br>[4.61] | 0.41<br>[4.74] | 0.33<br>[3.92] | 0.35<br>[4.07] |
|                                                                       | (5)           | 0.51<br>[2.59] | 0.46<br>[2.42] | 0.61<br>[3.46] | 0.73<br>[4.30] | 0.73<br>[3.93] | 0.22<br>[2.35]                                     | 0.26<br>[2.76]  | 0.26<br>[2.84] | 0.29<br>[3.05] | 0.14<br>[1.48] | 0.17<br>[1.81] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | BDC Quintiles |                |                |                |                | BDC Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 352            | 351            | 351            | 353            | 357            | 24                                                 | 24              | 25             | 28             | 32             |                |
|                                                                       | (2)           | 98             | 98             | 97             | 98             | 98             | 50                                                 | 50              | 51             | 52             | 51             |                |
|                                                                       | (3)           | 70             | 71             | 70             | 71             | 71             | 90                                                 | 92              | 93             | 95             | 94             |                |
|                                                                       | (4)           | 61             | 61             | 61             | 61             | 61             | 202                                                | 209             | 210            | 221            | 218            |                |
| (5)                                                                   | 56            | 56             | 56             | 56             | 56             | 1368           | 1278                                               | 1459            | 1699           | 2594           |                |                |



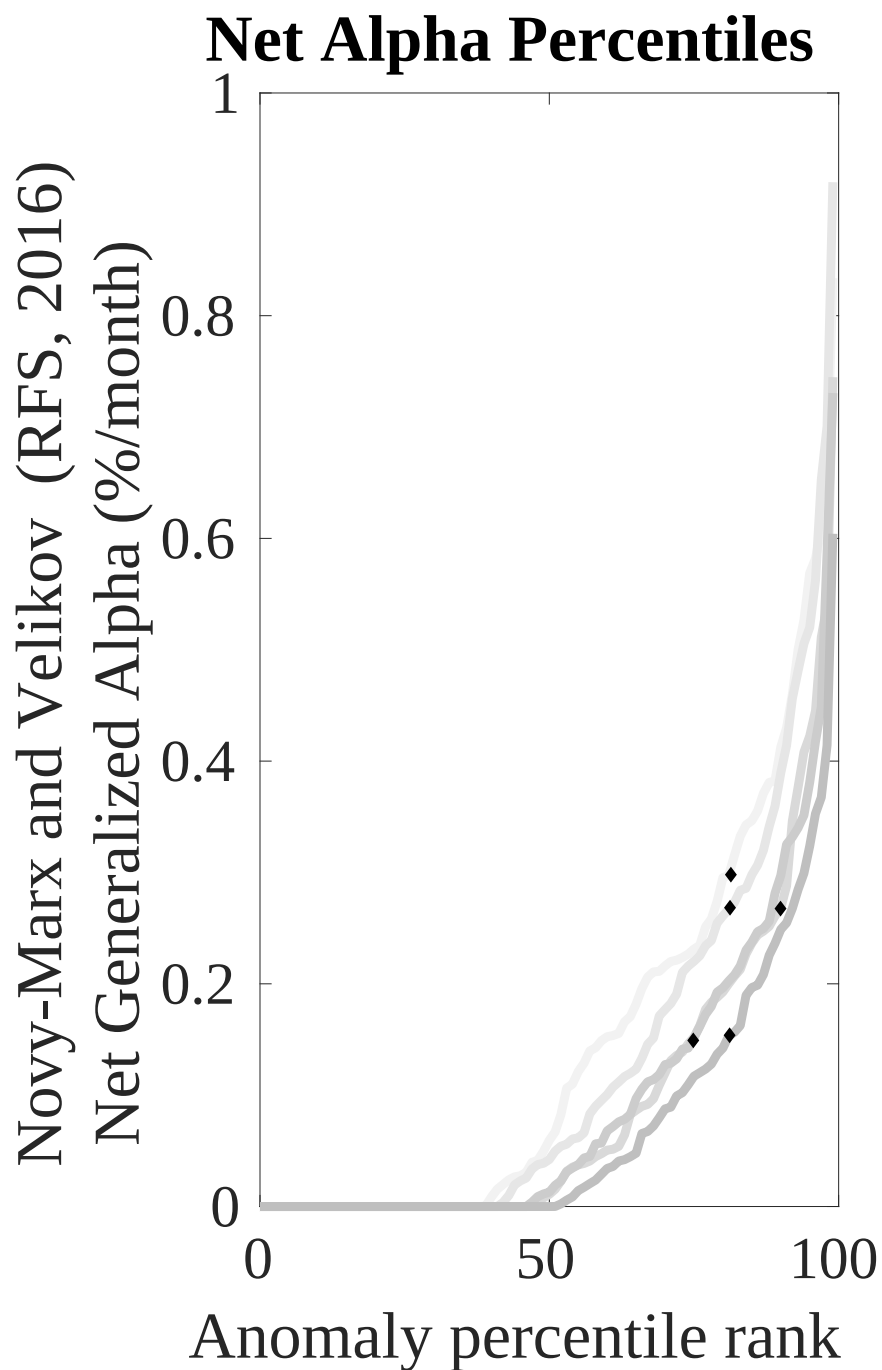
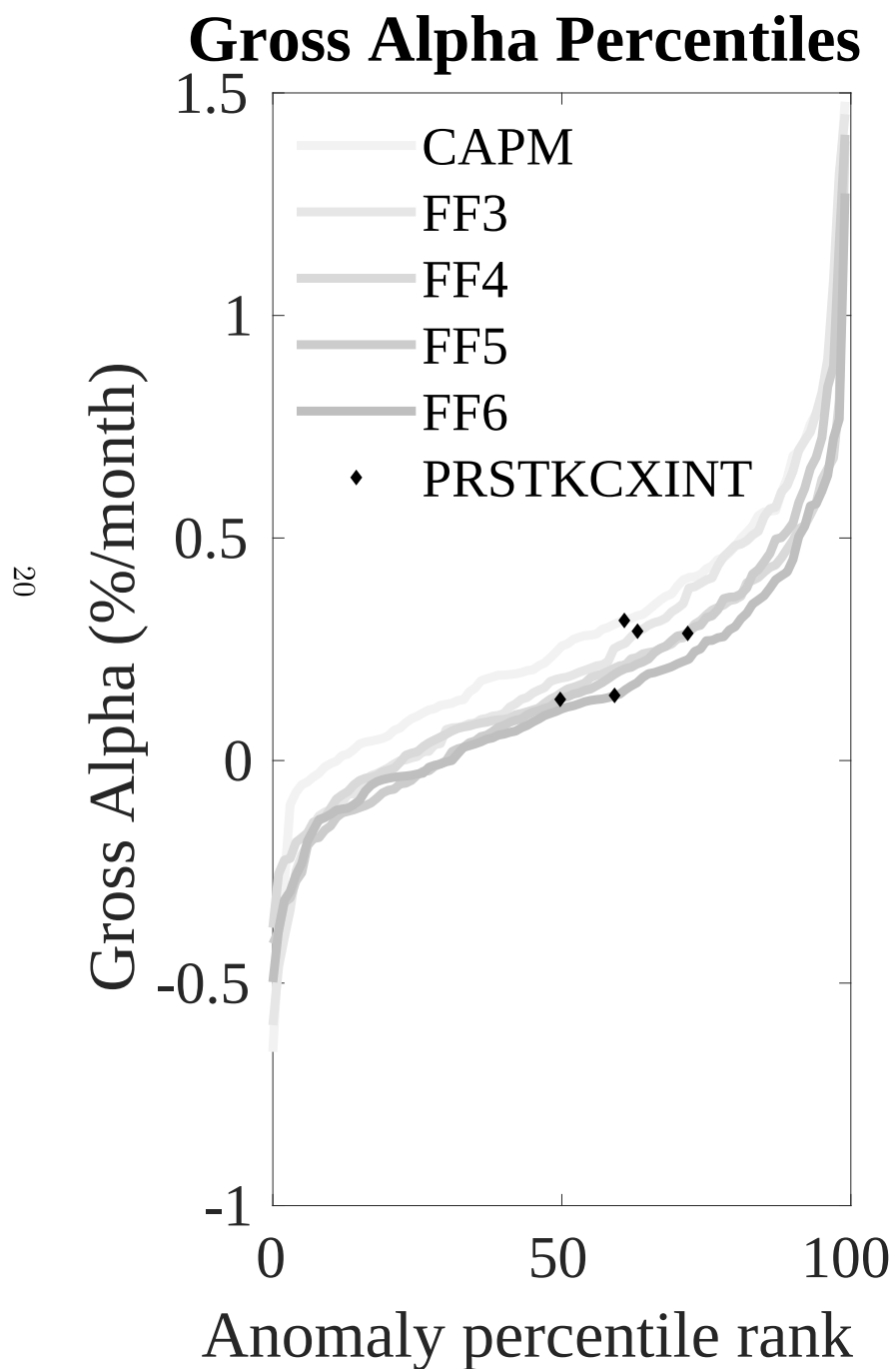
**Figure 2:** Distribution of Sharpe ratios.

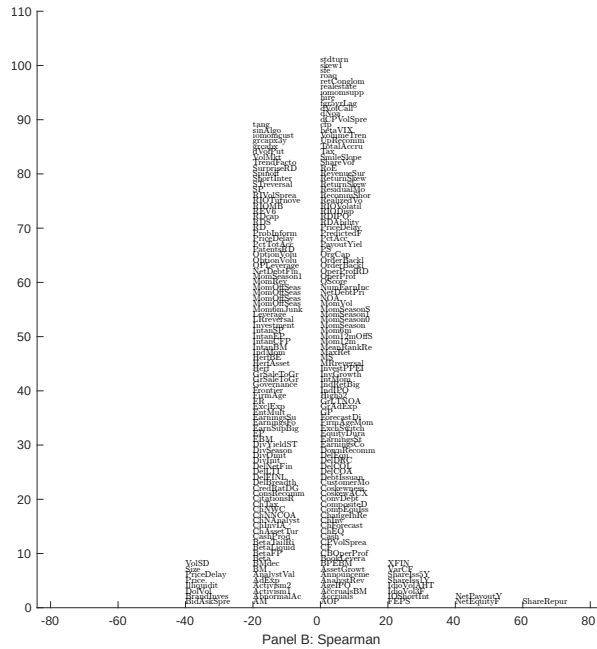
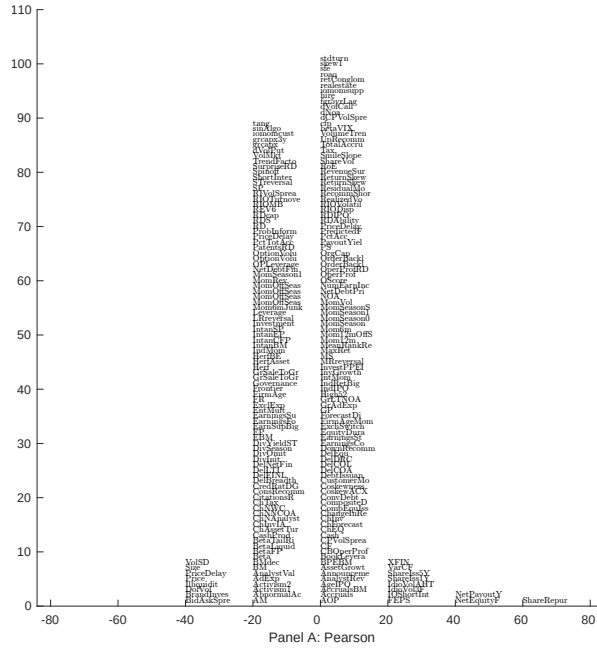
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

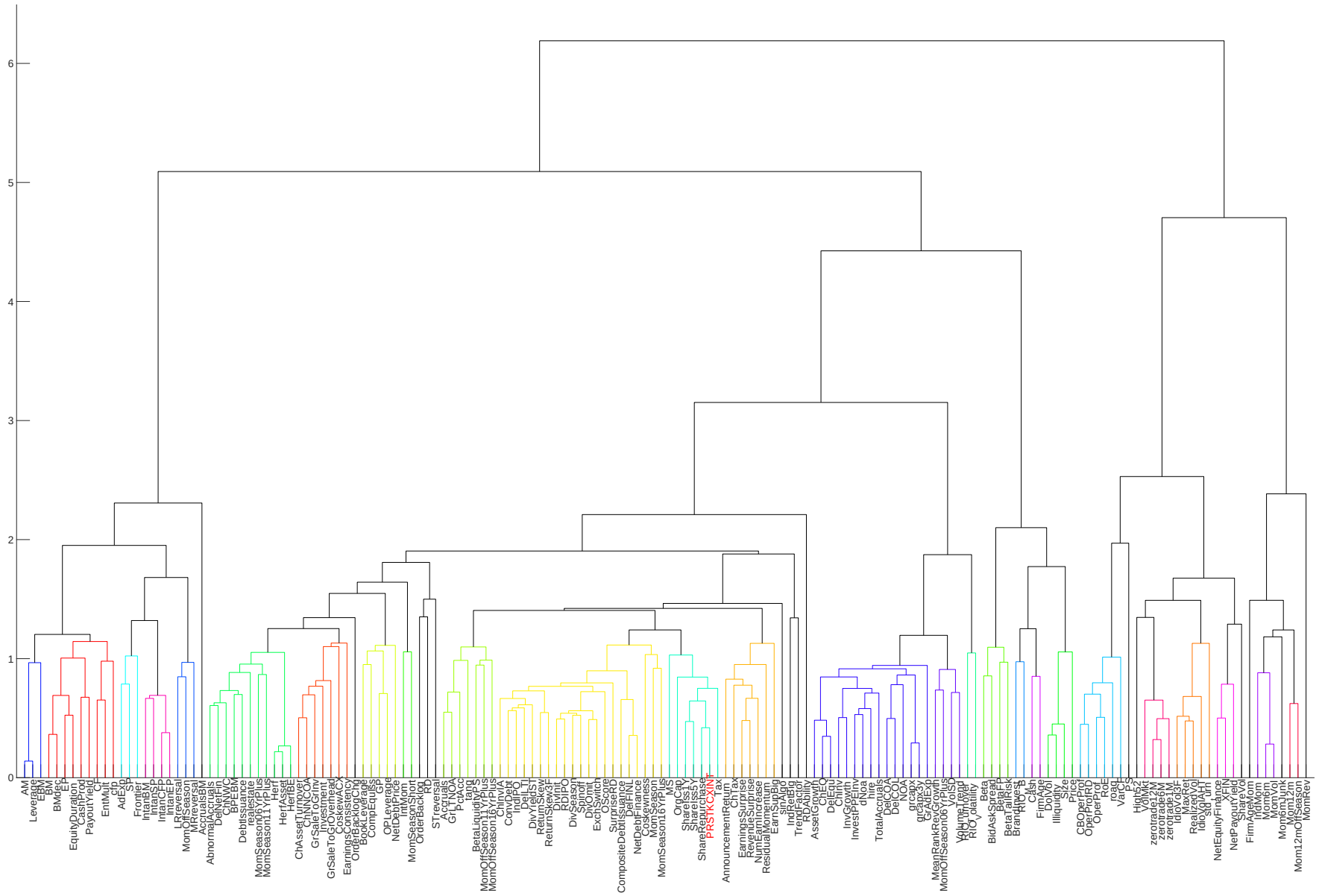
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.





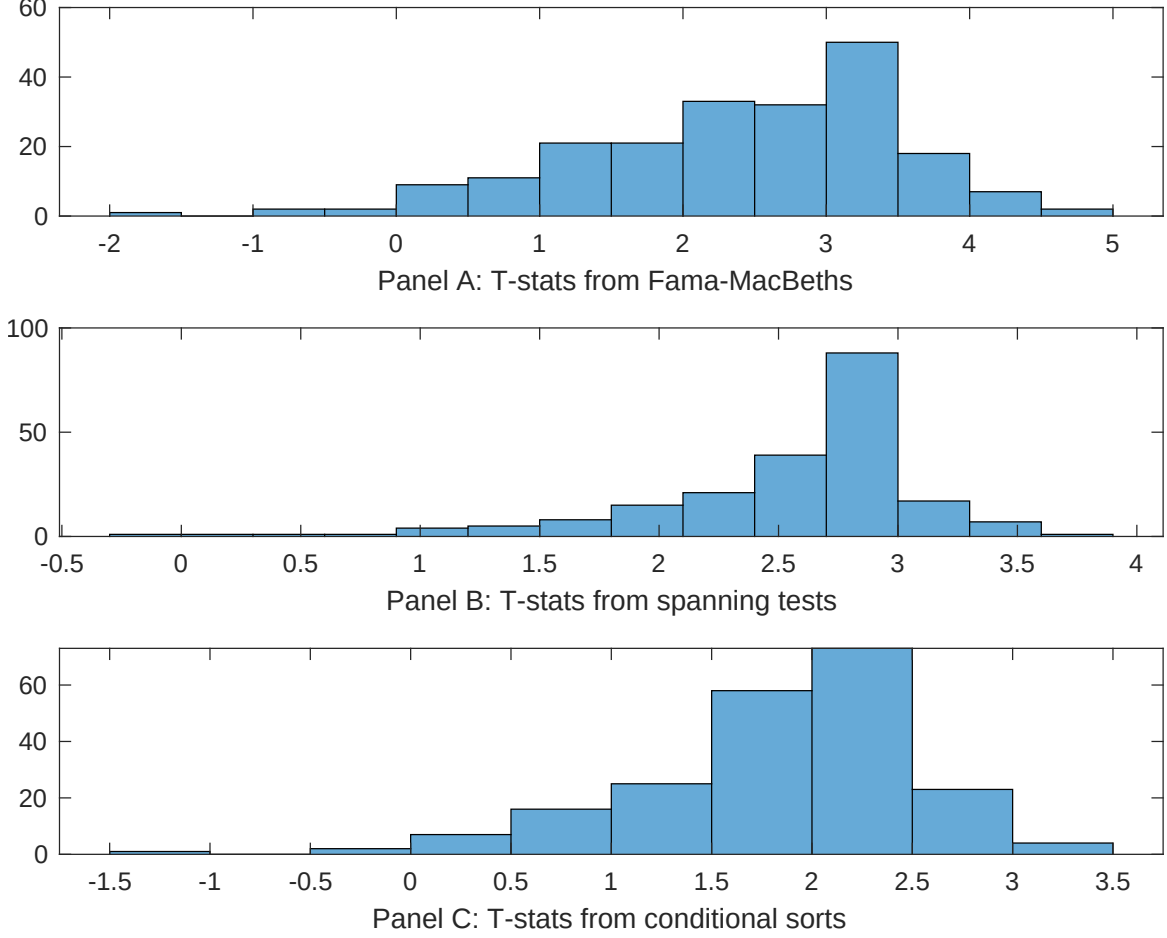
**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with BDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{BDC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BDC. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{BDC}BDC_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

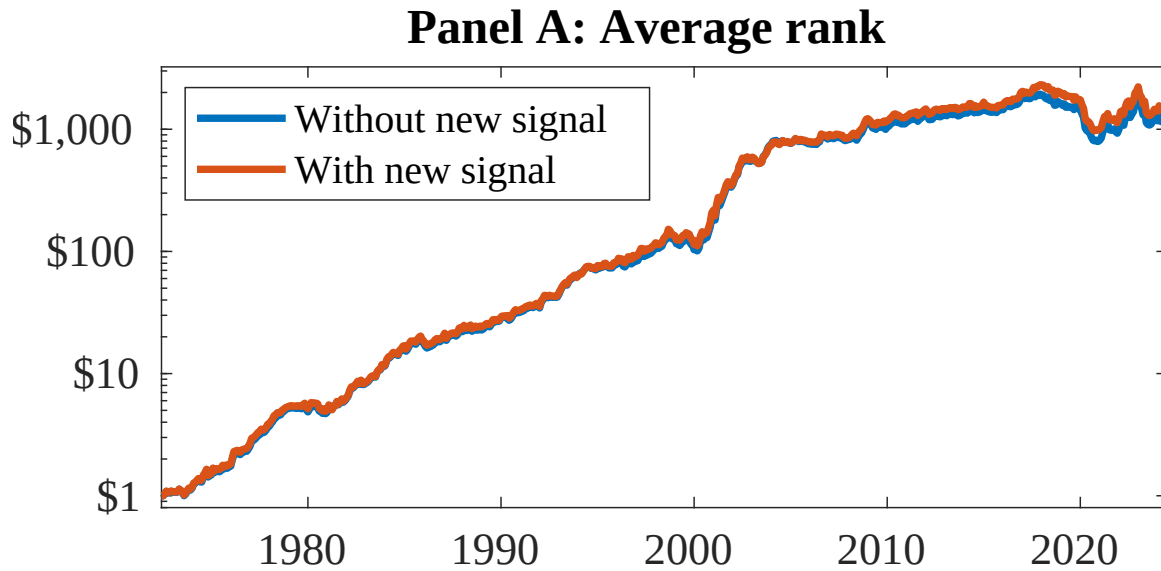
|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.10<br>[3.78] | 0.13<br>[5.17] | 0.13<br>[5.34] | 0.12<br>[5.09] | 0.13<br>[5.24] | 0.14<br>[5.94] | 0.13<br>[5.80]   |
| BDC             | 0.44<br>[1.12] | 0.78<br>[1.67] | 0.89<br>[2.16] | 0.60<br>[1.39] | 0.43<br>[1.12] | 0.12<br>[2.35] | 0.38<br>[0.67]   |
| Anomaly 1       | 0.25<br>[2.79] |                |                |                |                |                | 0.48<br>[0.96]   |
| Anomaly 2       |                | 0.20<br>[3.28] |                |                |                |                | -0.28<br>[-3.15] |
| Anomaly 3       |                |                | 0.18<br>[9.04] |                |                |                | 0.55<br>[2.54]   |
| Anomaly 4       |                |                |                | 0.29<br>[5.33] |                |                | 0.13<br>[2.40]   |
| Anomaly 5       |                |                |                |                | 0.21<br>[7.70] |                | 0.14<br>[5.85]   |
| Anomaly 6       |                |                |                |                |                | 0.34<br>[7.54] | 0.28<br>[6.08]   |
| # months        | 624            | 624            | 619            | 619            | 624            | 619            | 619              |
| $\bar{R}^2(\%)$ | 0              | 1              | 0              | 1              | 1              | 0              | 0                |



**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{BDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

|                 |                   |                  |                   |                   |                  |                   |                   |
|-----------------|-------------------|------------------|-------------------|-------------------|------------------|-------------------|-------------------|
| Intercept       | 0.18<br>[2.93]    | 0.18<br>[2.65]   | 0.13<br>[1.90]    | 0.17<br>[2.30]    | 0.12<br>[1.69]   | 0.08<br>[1.10]    | 0.16<br>[2.82]    |
| Anomaly 1       | 58.99<br>[18.24]  |                  |                   |                   |                  |                   | 44.67<br>[12.91]  |
| Anomaly 2       |                   | 43.57<br>[14.50] |                   |                   |                  |                   | 17.08<br>[4.37]   |
| Anomaly 3       |                   |                  | 38.10<br>[10.64]  |                   |                  |                   | 5.47<br>[1.40]    |
| Anomaly 4       |                   |                  |                   | 26.15<br>[9.14]   |                  |                   | 4.09<br>[1.44]    |
| Anomaly 5       |                   |                  |                   |                   | 33.65<br>[9.25]  |                   | 10.39<br>[2.97]   |
| Anomaly 6       |                   |                  |                   |                   |                  | 28.02<br>[6.93]   | -1.60<br>[-0.42]  |
| mkt             | -2.29<br>[-1.59]  | 4.37<br>[2.68]   | -2.45<br>[-1.50]  | 1.25<br>[0.71]    | 1.67<br>[0.95]   | -2.18<br>[-1.27]  | 2.63<br>[1.79]    |
| smb             | -8.61<br>[-3.88]  | -0.82<br>[-0.32] | -16.15<br>[-6.57] | -12.75<br>[-4.97] | -7.20<br>[-2.60] | -16.20<br>[-6.28] | 0.00<br>[0.00]    |
| hml             | -10.93<br>[-3.92] | -8.76<br>[-2.94] | -5.62<br>[-1.79]  | -15.07<br>[-4.44] | -0.72<br>[-0.22] | -2.62<br>[-0.80]  | -12.07<br>[-4.28] |
| rmw             | 9.56<br>[3.20]    | 1.21<br>[0.34]   | 4.77<br>[1.24]    | 10.86<br>[2.91]   | 7.81<br>[1.98]   | 16.06<br>[4.31]   | -8.45<br>[-2.44]  |
| cma             | 12.67<br>[2.98]   | -0.54<br>[-0.11] | 4.11<br>[0.79]    | 5.76<br>[1.08]    | 3.78<br>[0.70]   | 14.19<br>[2.69]   | -5.89<br>[-1.28]  |
| umd             | -1.60<br>[-1.12]  | -2.31<br>[-1.51] | -3.22<br>[-1.98]  | 0.07<br>[0.04]    | -1.79<br>[-1.08] | -3.05<br>[-1.78]  | -1.96<br>[-1.45]  |
| # months        | 624               | 624              | 620               | 620               | 624              | 620               | 620               |
| $\bar{R}^2(\%)$ | 53                | 46               | 40                | 37                | 37               | 34                | 60                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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